

Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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DATA STRUCTURES AND ALGORITHMS - III

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No: 01 **Team No:** 05

Project Title: Automated Student-to-Laboratory Allocation System Using Maximum Bipartite Matching

Abstract

Laboratory allocation is an important scheduling task in educational institutions, particularly when many students must be assigned to a limited number of laboratories according to their preferences and availability. This project develops an automated student-to-laboratory allocation system using maximum bipartite matching. Students and laboratories are represented as two disjoint vertex sets, with an edge representing a valid preference or possible allocation. The system constructs the bipartite graph from student, laboratory, and preference inputs and applies an augmenting-path based maximum matching algorithm to obtain the largest possible set of valid assignments. The approach prevents duplicate student allocations and assignment conflicts while maximizing the number of successful matches. The system generates a readable allocation report showing student-to-laboratory mappings, unmatched students, and allocation statistics. The project demonstrates the practical application of graph theory, maximum matching, algorithmic problem solving, optimization, and automated resource allocation in an educational environment, with scope for extensions such as laboratory capacities, weighted preferences, eligibility conditions, and time-slot availability.

Team Details

Roll Number(s)	Name of the Student	Signature
2520030428	Ritwik Kolluri	
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DSA Concepts / Algorithms Used

- Data Structures Used: Bipartite Graph, Adjacency List, Matching Arrays
- Algorithms / Techniques Used: Maximum Bipartite Matching using Augmenting Paths
- Core DSA Concept / Problem Addressed: Graph-based resource allocation and preference matching
- Time Complexity: $O(VE)$

- Space Complexity: $O(V + E)$

Guide Name and Signature (remarks if any)

Guide Name: CH Anuradha **Signature:** _____

Remarks: _____